Feature Reduction in Time Series - Intermediate Results

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Introduction

Feature Reduction:

Process of reducing the number of input in a dataset while retaining essential information.

Why perform Feature Reduction ?

Overview

Step 1

Data Preprocessing

Step 2

Feature Reduction using PCA

Step 3

Predictive Modelling using LSTM

Step 4

Evaluation

CNC Dataset

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Machines: CMX & DMC
Materials: Al & S
Components Produced: CP1 & CP2

Dataset

56 features (from original 92)

Features

Load
Torque / Velocity
Encoder Position
Command Speed
Control Difference / Position

Axes Covered

X (1)
Y (2)
Z (3)
Spindle (6)

Target Features

Current consumption for each axis
Ex: Current|1, Current|2 …

Dataset

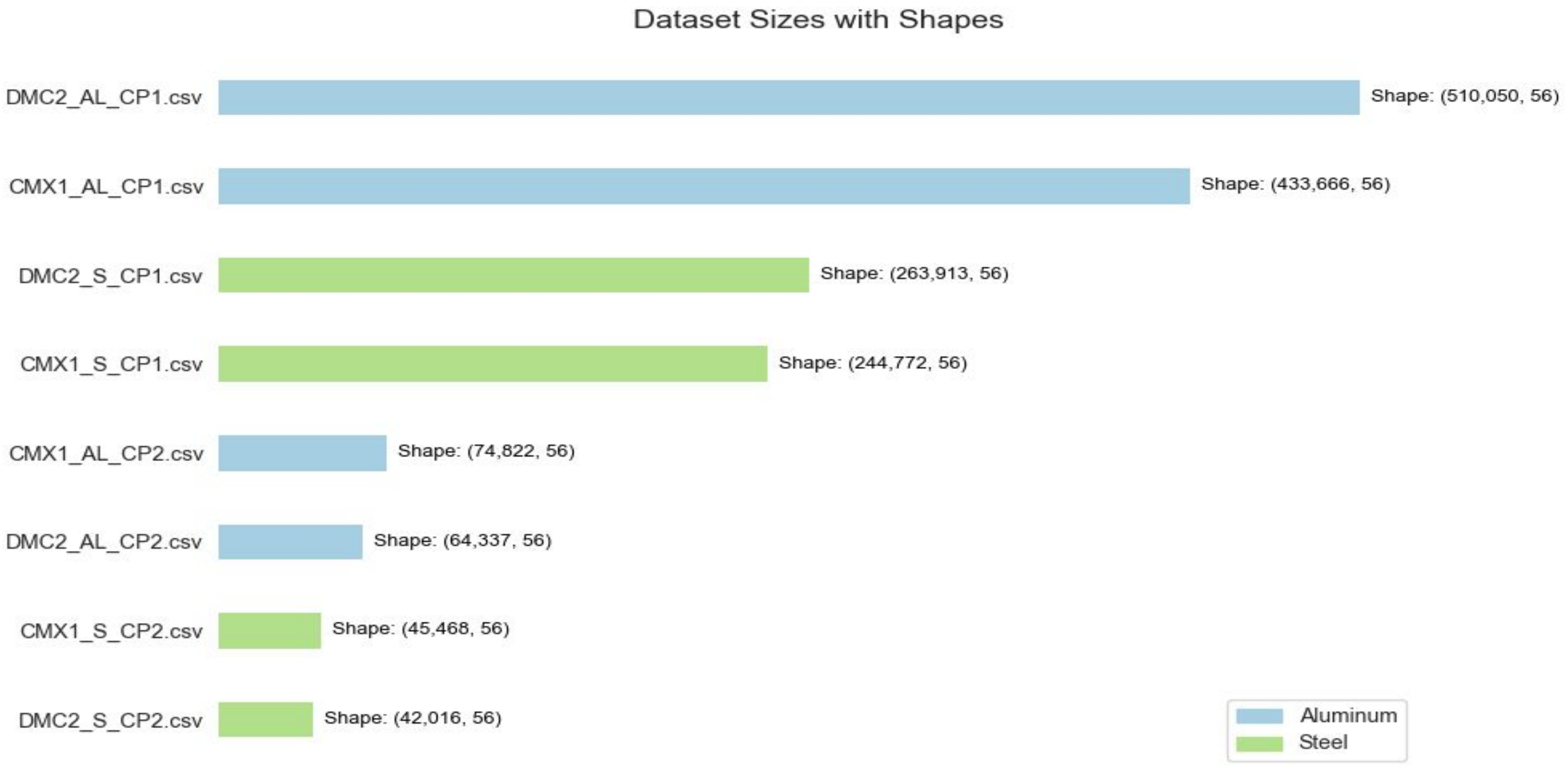


Fig 1 : Dimensional Comparison of 8 Datasets

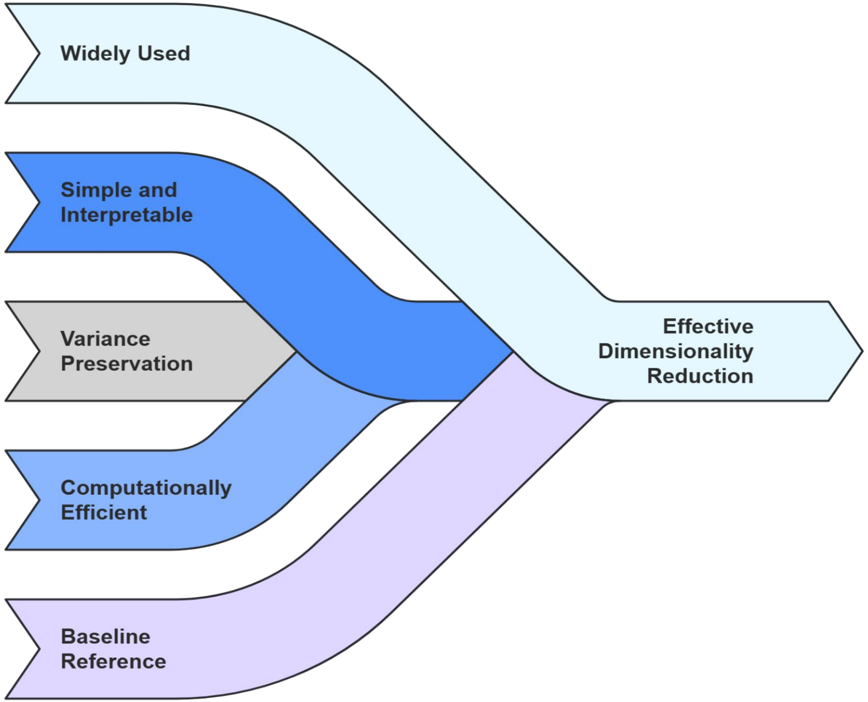
Architecture Selection

PCA (Principal Component Analysis)

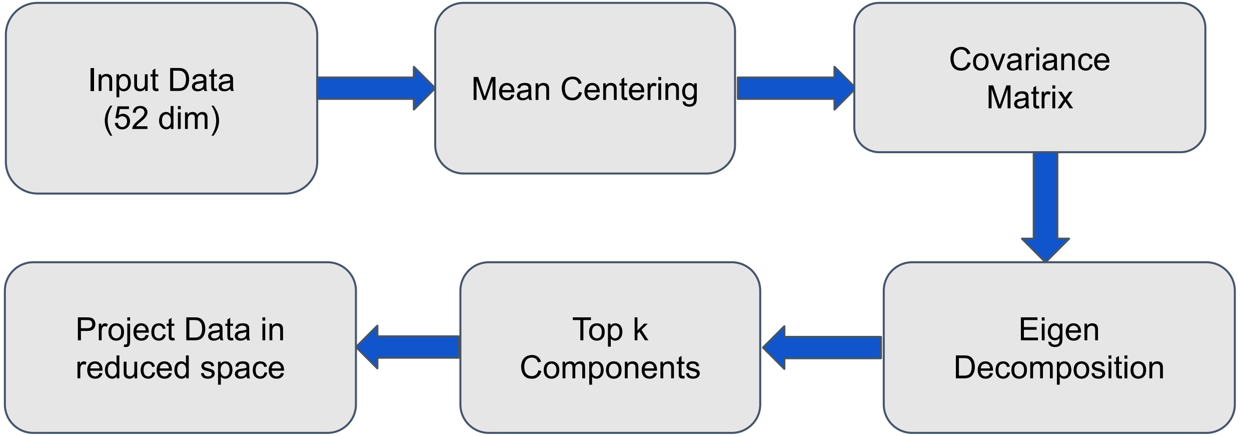
VAE (Variational Autoencoders)

Hybrid Autoencoders

Motivation for using PCA First



PCA Architecture

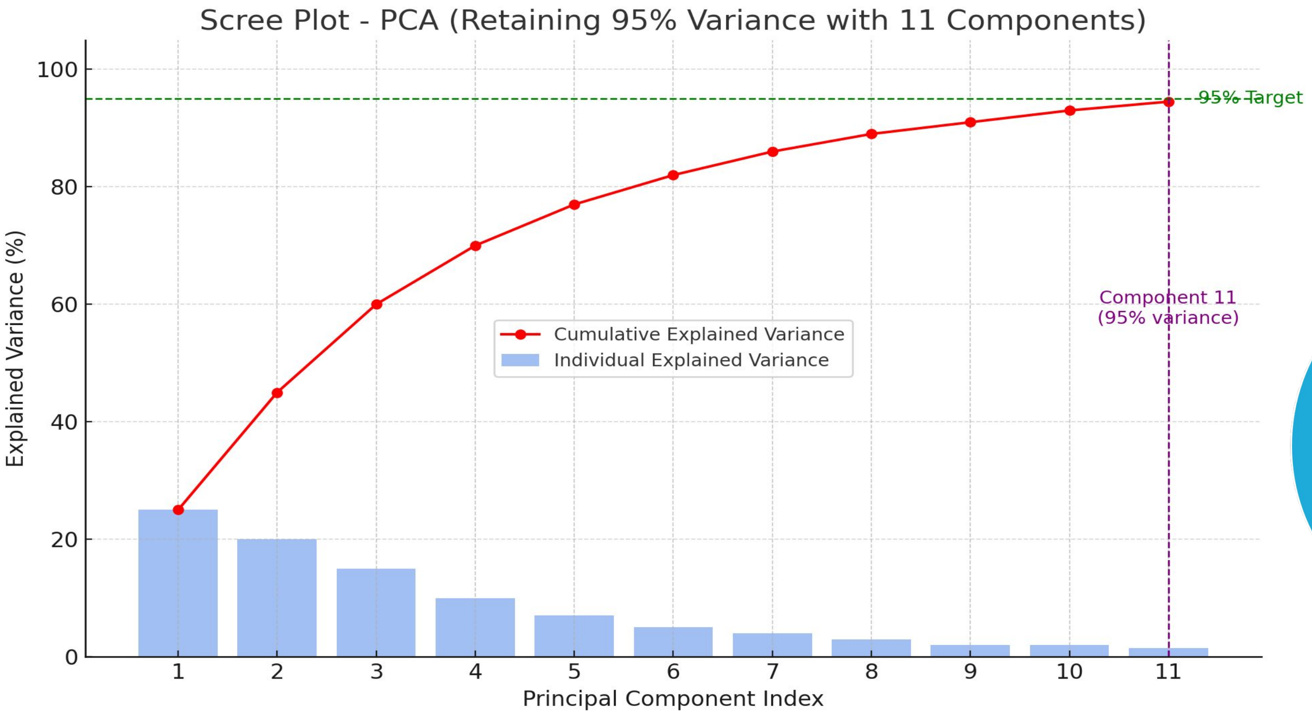


- **Mean Centering:** $X_{centered} = X - \bar{X}$
- **Covariance Matrix:** $\Sigma = \frac{1}{n-1}(X_{centered})^T X_{centered}$
- **Eigen Decomposition:** $\Sigma v = \lambda v$
- **Projection:** $Z = X_{centered} W$

Fig 2 : PCA Methodology

Symbol	Description
X	Input data matrix
Σ	Covariance matrix
v, λ	Eigenvectors and eigenvalues
W	Projection matrix (top k eigenvectors)
Z	Projected data

Explained Variance Across Principal Components



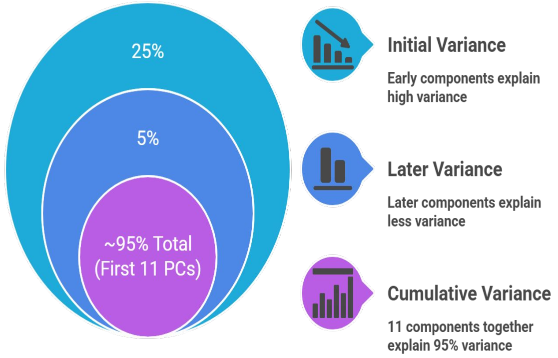


Fig 3 : Scree Plot

Loading Plot – Feature Contributions

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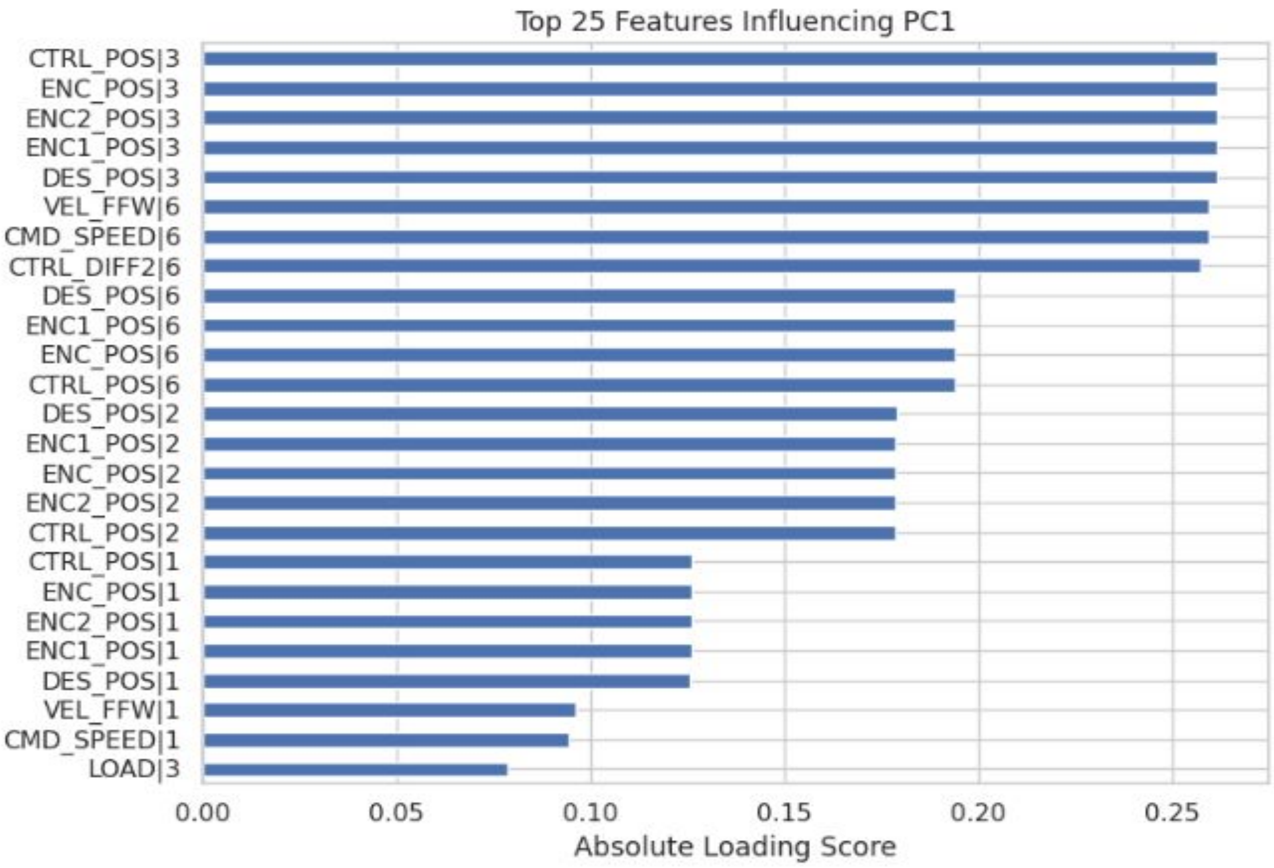


Fig 4 : Loading Plot for PC1

- **Top 25 features with the highest contributions** to Principal Component 1
- **Loading scores** = each feature's influence on the component
- Highlights which features drive most of the variance.
- Supports **feature understanding and model interpretation**

PCA Performance Evaluation

Data Preparation

Original Features = 52
PCA-reduced Feature = 11
Target = Current|6

LSTM on PCA-red. Data

Predict Target
Evaluate using MAE, RMSE, R^2

LSTM on Original Data

Predict Target
Evaluate using MAE, RMSE, R^2

Evaluation

Compare the Results

Evaluation Result

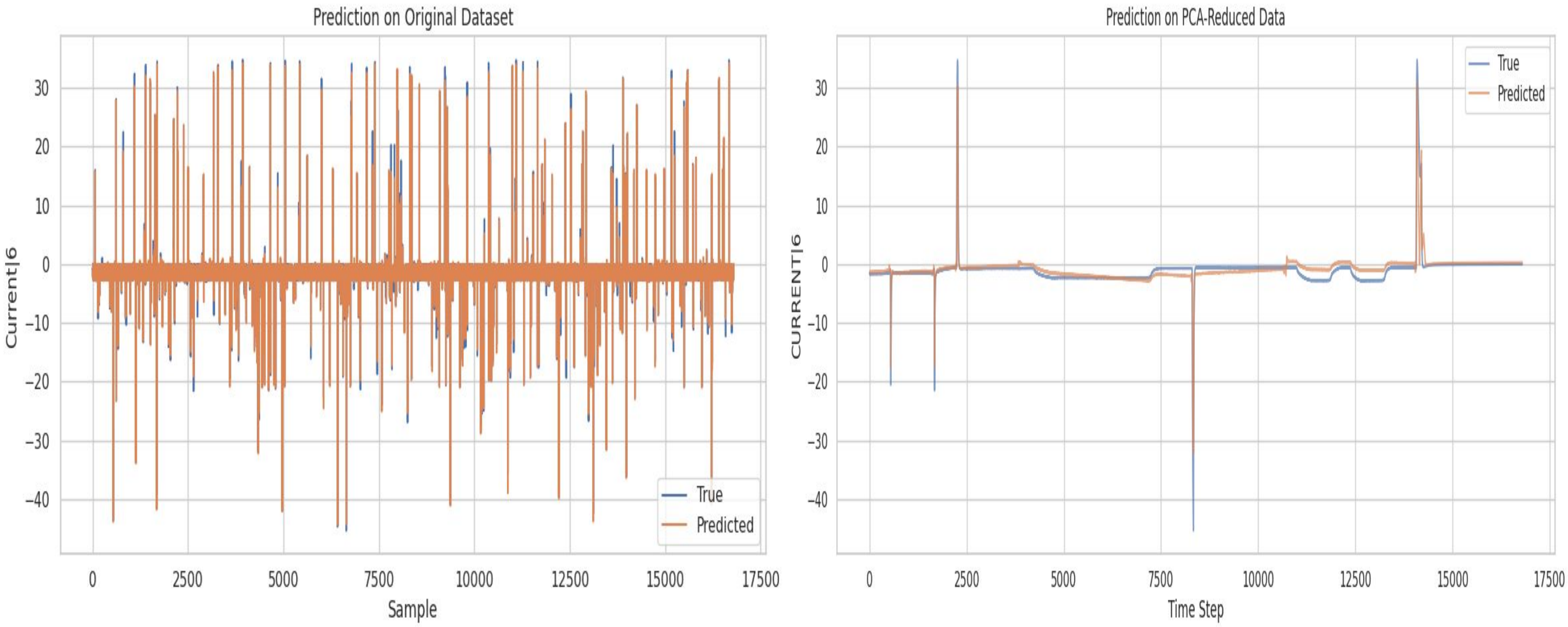


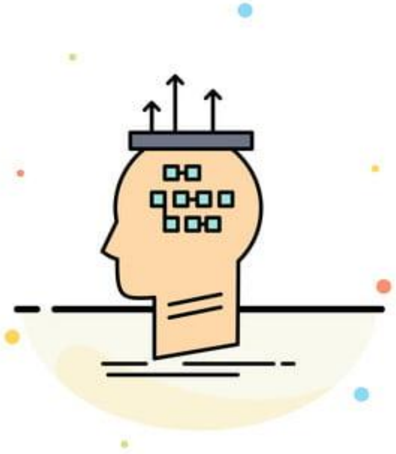
Fig 5 : Comparison between original dataset and PCA-red. dataset

Evaluation Result

Model	Features	MAE (Mean Absolute Error)	RMSE (Root Mean Square Error)	R^2 Score
LSTM + PCA red. Dataset	11	1.2817	2.2132	0.5521
LSTM + PCA red. Dataset	14	0.7468	1.3460	**0.8344**
LSTM + Original Dataset	52	0.0013	0.0022	**0.9959**

Table 1 : Evaluation Result

Conclusion



- PCA helped identify **top contributing features :** CTRL_DIFF2|2/1, CTRL_DIFF|2/1, CONT_DEV|2/1/3, CTRL_DIFF2|3, CTRL_DIFF|3, TORQUE|3

- **Least contributing features :** ENC2_POS|6, TORQUE_FFW|1/6/3/2 , CONT_DEV|6, CTRL_DIFF|6, LOAD|6, CTRL_POS|3, ENC_POS|3

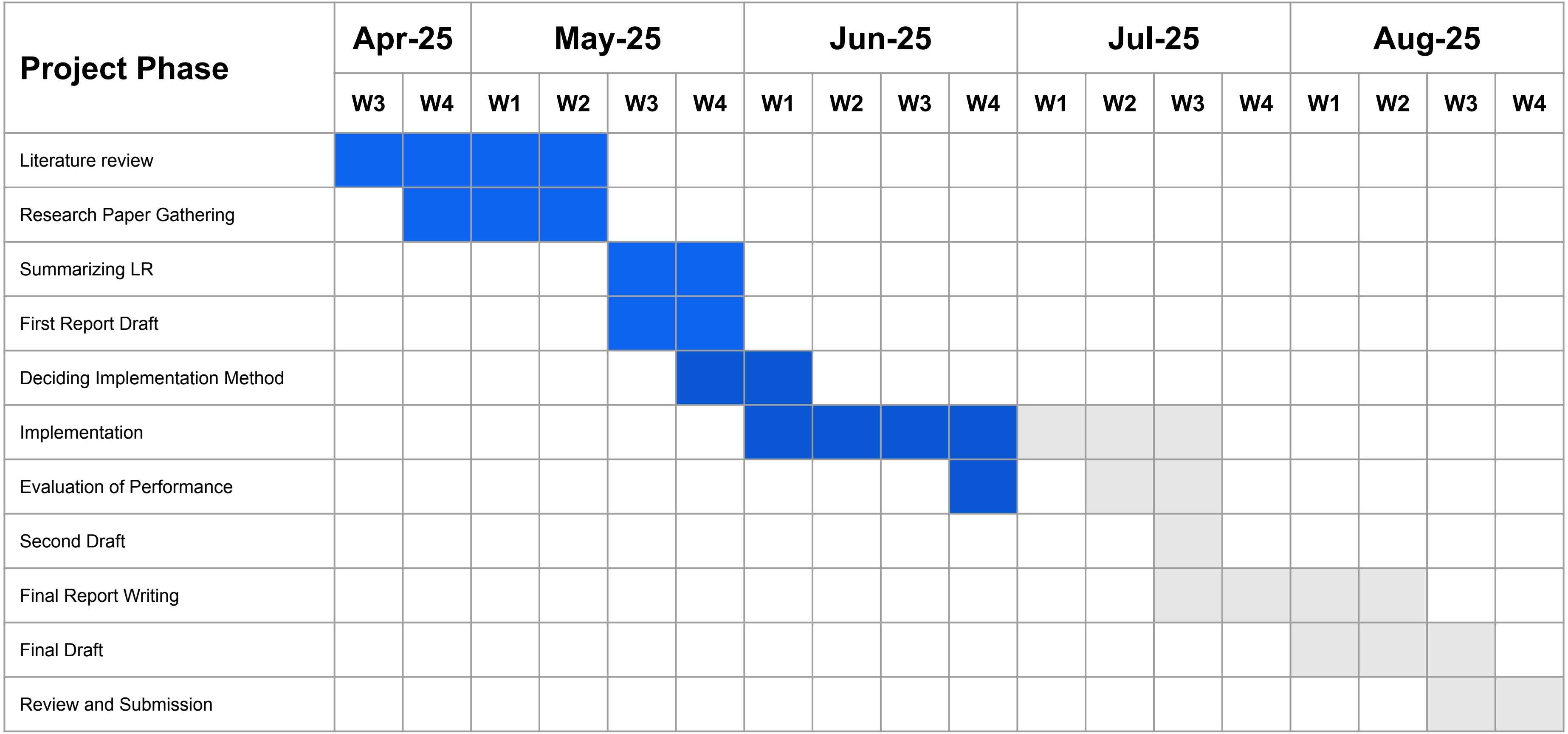
- **Trade-Off Observed** :
  - PCA reduces computation and model complexity
  - But, may lead to slight performance loss

- PCA with **14 components** offers a strong balance between **model simplicity** and **predictive power.**

Future Work

- Extend PCA analysis with **non-linear methods** like **VAE** and **Hybrid Autoencoders.**
- Improve generalization and reduce feature space without compromising **accuracy**.
- Identify **redundant vs. influential features** to enhance decision-making and model transparency.
- Incorporate **explainable AI (XAI)** techniques to understand which original features drive predictions.

Project Timeline



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Thank You For Your Attention!

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